Domain Specific Language and variables for systematic approach to genetic variant curation and interpretation

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Deep Learning-Guided Program Synthesis in a Specialized DSL is more effective than conventional modelling.

DSL and Annotation Store for Clinicians

Clinicians analyzing genomic data face three critical questions:

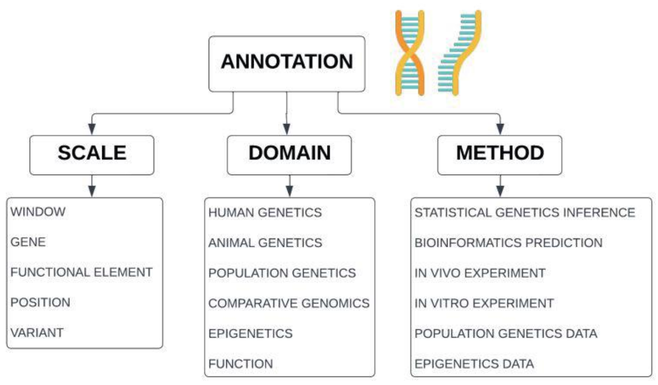
- Is the variant present in the patient?
- Does it affect the phenotype?
- What evidence links the variant to the phenotype?

Answers are based on annotations encompassing provenance, confidence, and relevant biological and clinical evidence.
Annotation Store — a system that categorizes annotations by genomic scale, resolution, knowledge domain, and acquisition method, making them more accessible and actionable.

The DSL represents rules as sequences of inclusion and exclusion criteria, facilitating the creation of lineage graphs that visually map the evidence for candidate variants.

The use of modern generative AI to write DSL scripts offers a transparent alternative to opaque AI models in genomic analysis. Unlike traditional models, DSL scripts are easily reviewable and approvable, making them ideal for peer and regulatory scrutiny while benefiting from the sophistication of AI-generated code.

Studies (e.g., https://arxiv.org/html/2412.04604v2) proved that for sophisticated tasks Deep Learning-Guided Program Synthesis (induction) is more effective than test-time training (TTT), also known as test-time fine-tuning (TTFT) (transduction).

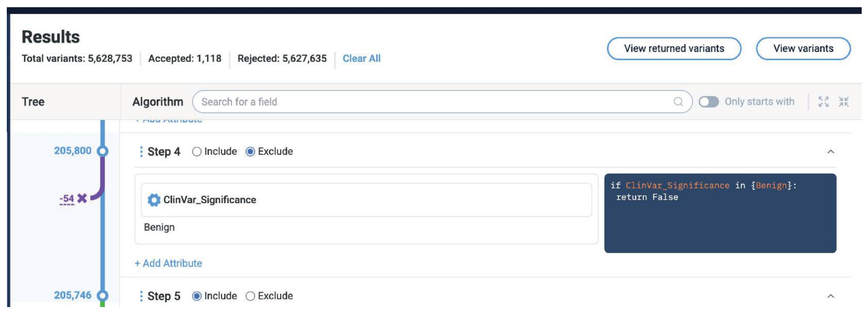


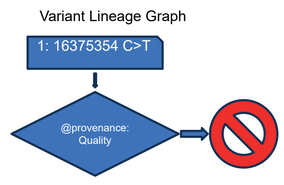
Filtration approach based on manually curated decision trees

- Sequence of inclusion and exclusion criteria instead of unreadable and unmaintainable nested if-then-else blocks, when nesting level approaches 5 and more
- Easy to visualize
- Drastically improves expalinability every step

```
if (
  (
    r in database 'DB1' with labels 'D' or 'DM'
  )
  OR (
    (r is rare /*'AF' < 0.01*/)
    AND (r is coding)
    AND (
      (
        r in database 'DB2' with labels 'V' or 'V1' or 'V2'
        AND (r is not in database 'DB2' with labels 'B')
      )
      OR (
        r in database 'DB2' with labels 'D' or 'DD'
      )
    )
  )
  OR (
    (r is very rare /* AF < 0.0007 */)
    AND (r is coding)
    AND (r is not in databases 'DB1' or 'DB2')
    AND (r is in database 'DB3' with label 'D')
  )
)
 then include
else
 exclude
```

```
if (r in database 'DB1' with labels 'D' or 'DM')
   then include
if (r is not rare /* AF >= 0.01 */)
   then exclude
if (r is not coding)
   then exclude
if (r in database 'DB2' with labels 'D' or 'DD')
   then include
if (r is in database 'DB2' with labels 'B')
   then exclude
if (r in database 'DB2' with labels 'V' or 'V1' or 'V2')
   then include
if (r is not very rare /* AF >= 0.0007 */)
   then exclude
if (r is in databases 'DB1' or 'DB2')
   then exclude
if (r is in database 'DB3' with label 'D')
   then include
exclude
```





Validation through annotations:

```
@purpose(evidence)
@knowledge_domain("Human Genetics")
@scale(variant)
@method("Clinical Evidence")
if (r in database 'DB1' with labels 'D' or 'DM')
   then include
```

We are refining the semantic and syntactic frameworks of DSLs, standardizing their use in variant curation, and expanding their potential, and regulatory compliance.

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